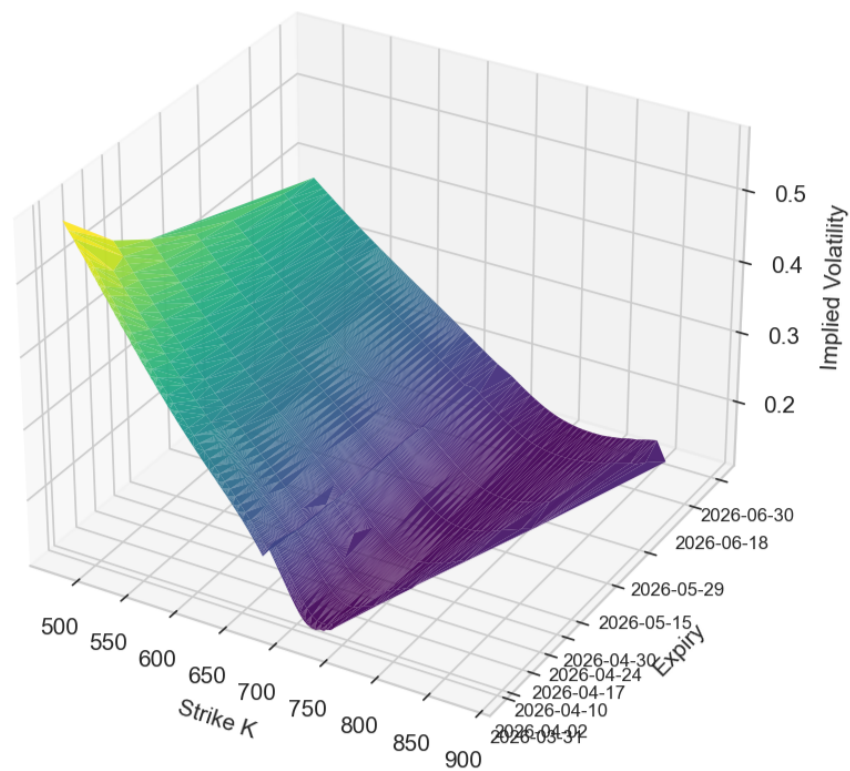


The Heston Model

Theory, Numerical Simulation and Calibration

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This seminar paper is based on the Heston model. The aim is not only to write down the model, but also to explain why the variance process has to be treated carefully. We first recall the pricing idea behind Black–Scholes. After that we introduce the Heston model and look at the CIR variance process. The important point is that $\sqrt{V_t}$ is not globally Lipschitz at zero. So the usual standard theorem for SDEs cannot simply be used. Later we use this theory for the numerical simulation, for Monte Carlo pricing and for the calibration to SPY option data. In the end we compare the model smile with the market smile and explain where the model still has visible errors.

Note on the Use of AI Tools. During the preparation of this paper, AI-assisted tools were used in a supporting role. ChatGPT was used for the linguistic revision of individual formulations and for translations. The code used to generate the numerical plots was created with the help of Codex and was subsequently reviewed, adapted, and validated.

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1 Introduction

The Heston model is a stochastic volatility model. Before we use it, we first recall the basic idea of arbitrage-free pricing. In the Black–Scholes model the payoff of a European option is a function of the terminal stock value. For example, for a European call we have

$$f(S_T) = (S_T - K)^+ := \max(S_T - K, 0).$$

Under the physical measure P , the stock price is written as

$$dS_t = \mu S_t dt + \sigma S_t dB_t.$$

Here μ is the historical drift and σ is the constant volatility. As second tradable asset we use the bank account

$$B_t^0 = e^{rt}, \quad dB_t^0 = rB_t^0 dt, \quad B_0^0 = 1.$$

So the model contains the stock and the risk-free asset. Under the usual assumptions of a frictionless market one can replicate the option payoff. Therefore the price is not computed with the historical drift μ , but under a risk-neutral measure $\mathbb{Q}$. Under this measure we have

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}, \quad C_0 = e^{-rT} \mathbb{E}_{\mathbb{Q}}[f(S_T)].$$

Thus the discounted stock price is a martingale. This is the basic pricing principle used in the rest of the paper.

Example 1.1. We now look at a very small one-period example. Let $S_0 = 100$. At the end of the period the stock takes the value 110 with probability 90% and the value 90 with probability 10% under the historical probabilities. For a call with strike $K = 100$ the terminal payoffs are therefore $C_u = 10$ and $C_d = 0$. We assume for simplicity that $r = 0$. A replicating portfolio with Δ shares and a bank position B then gives

$$110\Delta + B = 10, \quad 90\Delta + B = 0.$$

Solving this system gives $\Delta = \frac{1}{2}$ and $B = -45$. Hence the arbitrage-free price is

$$C_0 = \Delta S_0 + B = \frac{1}{2} \cdot 100 - 45 = 5.$$

If we used the historical probabilities directly, we would get

$$0.9 \cdot 10 + 0.1 \cdot 0 = 9.$$

This cannot be the arbitrage-free price. If the option were traded for 9, one could sell the option, build the replicating portfolio for 5, and keep the difference. So the example shows why the risk-neutral probability is the relevant one. From

$$100 = q \cdot 110 + (1 - q) \cdot 90$$

we get $q = \frac{1}{2}$, and therefore again

$$C_0 = q \cdot 10 + (1 - q) \cdot 0 = 5.$$

In applications we do not observe the volatility parameter directly. We observe market prices. Therefore we invert the Black–Scholes formula and search for the volatility σ_{impl} that reproduces the observed option price. This number is called implied volatility. If Black–Scholes were correct for all strikes and maturities, then this implied volatility would be constant. In market data this is not the case.

For SPY options from MarketData.app we plot the implied volatility against log-moneyness

$$\log(K/F), \quad F = S_0 e^{rT}.$$

Values $\log(K/F) > 0$ correspond to strikes above the forward price.

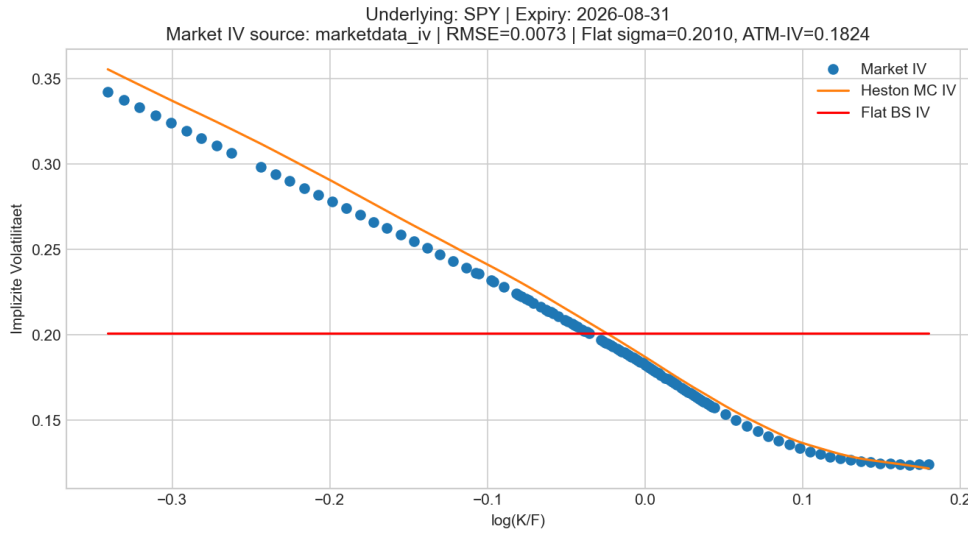


Figure 1: Out-of-sample smile for one SPY expiry: market IV and Heston fit in log-moneyness representation.

The figure shows the main problem. The implied volatility is not constant, but depends on moneyness. So a model with one constant volatility cannot describe the observed smile. The Heston model is one natural extension, because it makes the volatility random and allows it to return to a long-run level.

We now define the model that is used in this paper. Under the risk-neutral measure

$\mathbb{Q}$, the Heston model is given by

$$dS_t = rS_t dt + \sqrt{V_t}S_t dW_t^S, \quad (1)$$

$$dV_t = \kappa(\theta - V_t) dt + \xi\sqrt{V_t}dW_t^V, \quad (2)$$

where $\langle W^S, W^V \rangle_t = \rho t$, $S_0 > 0$, $V_0 \geq 0$, $\kappa, \theta, \xi > 0$, and $\rho \in (-1, 1)$. The parameter κ describes the speed of mean reversion, θ the long-run variance level, ξ the volatility of variance, and ρ the correlation between stock and variance shocks.

The paper proceeds as follows. First we define the model. Then we study the variance process mathematically. After that we discuss the numerical approximation and the Monte Carlo error. Finally we use the model for calibration and compare the resulting implied volatility smile with market data.

The difficult part is the variance equation. The coefficient $\sqrt{V_t}$ is not globally Lipschitz. So the usual SDE theory is not enough. This is why the CIR process is the main theoretical object in the next chapter.

2 Risk-Neutral Model Framework

We work on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{Q}),$$

which satisfies the usual conditions. On this space we have a two-dimensional Brownian motion

$$(W^S, W^\perp).$$

For a fixed correlation parameter $\rho \in (-1, 1)$, we define

$$W_t^V := \rho W_t^S + \sqrt{1 - \rho^2} W_t^\perp.$$

Then W^V is again a Brownian motion and

$$\langle W^S, W^V \rangle_t = \rho t.$$

So the two Brownian motions are coupled by ρ . If $\rho < 0$, then a downward stock move tends to come together with an upward variance move. This is exactly the typical skew effect observed in equity markets.

The Heston model now consists of two continuous adapted processes (S, V) , where

$S_t > 0$ and $V_t \geq 0$. They satisfy

$$S_t = S_0 + \int_0^t r S_s ds + \int_0^t \sqrt{V_s} S_s dW_s^S, \quad (3)$$

$$V_t = V_0 + \int_0^t \kappa(\theta - V_s) ds + \int_0^t \xi \sqrt{V_s} dW_s^V. \quad (4)$$

The first equation is not the main difficulty. Once the variance path is known, the stock equation behaves like a geometric Brownian motion with time-dependent volatility. The second equation is more delicate, because of the square root.

The Feller condition is

$$2\kappa\theta \geq \xi^2.$$

It separates two cases. If it holds and $V_0 > 0$, the boundary zero is not reached. If it is violated, the process can hit zero. This boundary behavior is important both for the theory and for the numerical scheme. Therefore we prove it carefully in the next chapter.

3 Mathematical Properties of the Variance Process (CIR)

We now consider the variance process in the Heston model. So we look at the SDE

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t, \quad V_0 \geq 0.$$

The important point is the square root. The function $x \mapsto \sqrt{x}$ is not globally Lipschitz at zero. Therefore we cannot simply use the standard existence and uniqueness theorem for SDEs with globally Lipschitz coefficients. We first introduce squared Bessel processes and then transform them into the CIR process. This is the same idea that is used in the standard literature; see for example [7].

3.1 Squared Bessel Processes

Definition 3.1 (Squared Bessel process). *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space and let W be a Brownian motion. A non-negative adapted process with continuous paths is called a squared Bessel process of dimension $\delta \geq 0$, written $\text{BESQ}(\delta)$, if*

$$X_t = X_0 + \delta t + 2 \int_0^t \sqrt{X_s} dW_s$$

holds almost surely for all $t \geq 0$.

For intuition we can set $R_t = \sqrt{X_t}$. On intervals where $X_t > 0$, Ito's formula formally

gives

$$dR_t = \frac{\delta - 1}{2R_t} dt + dW_t.$$

This formula explains the connection to the usual Bessel process. In the proof below we only need the squared process X .

Before proving the CIR result, we need two standard facts. They are quoted from [8, Chapter XI, Theorems 1.1 and 1.2].

Theorem 3.2 (Existence and uniqueness of $\text{BESQ}(\delta)$). *For every $\delta \geq 0$ and every initial value $X_0 \geq 0$, the $\text{BESQ}(\delta)$ SDE has a unique strong solution. The solution is non-negative, has continuous paths, and is a strong Markov process.*

Theorem 3.3 (Boundary classification of $\text{BESQ}(\delta)$). *Let $X \sim \text{BESQ}(\delta)$ with $X_0 > 0$, and let $\tau_0^X := \inf\{t \geq 0 : X_t = 0\}$. Then:*

- *If $\delta \geq 2$, then $\mathbb{P}(\tau_0^X < \infty) = 0$.*
- *If $0 < \delta < 2$, then $\mathbb{P}(\tau_0^X < \infty) = 1$.*
- *If $\delta = 0$, then zero is reached in finite time and is absorbing.*

Lemma 3.4 (Yamada–Watanabe). *If weak existence and pathwise uniqueness hold for an SDE, then the SDE has a unique strong solution.*

We will use this lemma only after we have constructed a weak CIR solution and shown pathwise uniqueness; see [9].

3.2 Existence and Uniqueness for the CIR Process

Theorem 3.5 (Existence and uniqueness of the CIR process). *Let $\kappa, \theta, \xi > 0$ and $V_0 \geq 0$. Then the CIR SDE*

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t$$

has a unique strong solution. In particular, $V_t \geq 0$ for all $t \geq 0$ almost surely.

Proof. We start with a squared Bessel process and then transform it. Set

$$\delta := \frac{4\kappa\theta}{\xi^2}, \quad \varphi(t) := \frac{\xi^2}{4\kappa}(e^{\kappa t} - 1).$$

Then φ is increasing and

$$\varphi'(t) = \frac{\xi^2}{4} e^{\kappa t}.$$

First we construct a weak solution. Choose a filtered probability space carrying a Brownian motion B and a $\text{BESQ}(\delta)$ process X with $X_0 = V_0$. We define

$$V_t := e^{-\kappa t} X_{\varphi(t)}, \quad \mathcal{G}_t := \mathcal{F}_{\varphi(t)}.$$

By the deterministic time-change theorem [A.1](#), $M_t := B_{\varphi(t)}$ is a continuous martingale with quadratic variation $\langle M \rangle_t = \varphi(t)$. Hence

$$dX_{\varphi(t)} = \delta\varphi'(t) dt + 2\sqrt{X_{\varphi(t)}} dM_t.$$

Using the product rule from Lemma [A.2](#), we get

$$dV_t = -\kappa V_t dt + e^{-\kappa t} \delta\varphi'(t) dt + 2e^{-\kappa t} \sqrt{X_{\varphi(t)}} dM_t.$$

Since $X_{\varphi(t)} = e^{\kappa t} V_t$, it follows that

$$\sqrt{X_{\varphi(t)}} = e^{\kappa t/2} \sqrt{V_t}.$$

Now define

$$\widetilde{W}_t := \frac{2}{\xi} \int_0^t e^{-\kappa s/2} dM_s.$$

Then

$$\langle \widetilde{W} \rangle_t = \frac{4}{\xi^2} \int_0^t e^{-\kappa s} d\langle M \rangle_s = \frac{4}{\xi^2} \int_0^t e^{-\kappa s} \varphi'(s) ds = t.$$

So Lemma [A.3](#) shows that $\widetilde{W}$ is a Brownian motion. For the drift we compute

$$e^{-\kappa t} \delta\varphi'(t) = e^{-\kappa t} \frac{4\kappa\theta}{\xi^2} \frac{\xi^2}{4} e^{\kappa t} = \kappa\theta.$$

Therefore

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} d\widetilde{W}_t.$$

So a weak solution has been constructed.

It remains to show pathwise uniqueness. Let V^1 and V^2 be two CIR solutions on the same probability space, driven by the same Brownian motion and with the same initial value. For $i = 1, 2$ set

$$Y_t^i := e^{\kappa t} V_t^i, \quad X_u^i := Y_{t(u)}^i, \quad t(u) := \varphi^{-1}(u) = \frac{1}{\kappa} \log\left(1 + \frac{4\kappa u}{\xi^2}\right).$$

The product rule gives

$$dY_t^i = \kappa\theta e^{\kappa t} dt + \xi e^{\kappa t/2} \sqrt{Y_t^i} dW_t.$$

After the inverse time change there is a Brownian motion $\widehat{W}$ such that

$$dW_{t(u)} = \sqrt{t'(u)} d\widehat{W}_u, \quad t'(u) = \frac{4}{\xi^2} e^{-\kappa t(u)}.$$

Thus for $i = 1, 2$

$$\begin{aligned} dX_u^i &= \kappa\theta e^{\kappa t(u)} t'(u) du + \xi e^{\kappa t(u)/2} \sqrt{X_u^i} dW_{t(u)} \\ &= \frac{4\kappa\theta}{\xi^2} du + 2\sqrt{X_u^i} d\widehat{W}_u = \delta du + 2\sqrt{X_u^i} d\widehat{W}_u. \end{aligned}$$

So X^1 and X^2 solve the same BESQ(δ) SDE with the same Brownian motion and the same initial value. By pathwise uniqueness for squared Bessel processes we get $X^1 = X^2$, and therefore also $V^1 = V^2$.

We now have weak existence and pathwise uniqueness. By Yamada–Watanabe this gives a unique strong solution. Finally, $V_t = e^{-\kappa t} X_{\varphi(t)}$ and $X \geq 0$, so $V_t \geq 0$ for all $t \geq 0$ almost surely. $\square$

3.3 Feller Condition for the CIR Process

Theorem 3.6 (Feller condition for CIR). *Let V be the CIR process with $V_0 > 0$, and let $\tau_0^V := \inf\{t \geq 0 : V_t = 0\}$. Then:*

- *If $2\kappa\theta \geq \xi^2$, then $\mathbb{P}(\tau_0^V < \infty) = 0$.*
- *If $2\kappa\theta < \xi^2$, then $\mathbb{P}(\tau_0^V < \infty) = 1$.*

Proof. From the construction above we have

$$V_t = e^{-\kappa t} X_{\varphi(t)}, \quad X \sim \text{BESQ}(\delta), \quad \delta = \frac{4\kappa\theta}{\xi^2}.$$

Since $e^{-\kappa t} > 0$,

$$V_t = 0 \iff X_{\varphi(t)} = 0.$$

Because φ is continuous, strictly increasing, and maps $[0, \infty)$ onto $[0, \infty)$, this gives

$$\tau_0^V = \varphi^{-1}(\tau_0^X), \quad \tau_0^X := \inf\{u \geq 0 : X_u = 0\}.$$

So

$$\tau_0^V < \infty \iff \tau_0^X < \infty.$$

Now we use the boundary classification of squared Bessel processes. It says

$$\mathbb{P}(\tau_0^X < \infty) = 0 \quad \text{if } \delta \geq 2,$$

and

$$\mathbb{P}(\tau_0^X < \infty) = 1 \quad \text{if } 0 < \delta < 2.$$

Since $\delta = 4\kappa\theta/\xi^2$, the condition $\delta \geq 2$ is the same as $2\kappa\theta \geq \xi^2$. Also $0 < \delta < 2$ is the same as $2\kappa\theta < \xi^2$. This proves the theorem. $\square$

Remark 3.7 (Interpretation of the Feller condition). *The Feller condition says whether the mean reversion is strong enough to keep the variance away from zero. If $2\kappa\theta \geq \xi^2$, the process does not hit zero. If $2\kappa\theta < \xi^2$, it hits zero with probability one. This matters numerically, because near zero the square-root term determines the behavior of the scheme.*

4 Convergence of Monte Carlo Option Prices under FTE and Log-Euler

4.1 Aim and Overview

After the CIR process has been treated theoretically, we now look at the numerical part. The aim is to approximate European option prices in the Heston model. Again the variance equation is the relevant difficulty. Because of the term $\sqrt{V_t}$, a simple Euler scheme may produce negative intermediate values. Then the square root would no longer be defined. Therefore we use the full-truncation Euler method. The convergence results used below are quoted from the literature and are only stated in the form needed for the later error estimate.

4.2 Full-Truncation Euler for the Variance Process

For the interpretation of the literature results, the Feller coefficient

$$\nu := \frac{2\kappa\theta}{\xi^2}$$

is central. It determines the boundary behavior of the CIR process and also appears in the assumptions of the convergence theorems. A naive Euler scheme can become negative. For this reason we truncate negative values at zero before evaluating drift and diffusion. This is the idea behind the full-truncation Euler method.

Let $N \in \mathbb{N}$, $\Delta := T/N$, and $t_n := n\Delta$. For the joint simulation of stock and variance we take independent standard normal variables $Z_n^{(1)}, Z_n^{(2)}$ and set

$$\Delta W_n^S := \sqrt{\Delta} Z_n^{(1)}, \quad \Delta W_n^V := \sqrt{\Delta} \left(\rho Z_n^{(1)} + \sqrt{1 - \rho^2} Z_n^{(2)} \right). \quad (5)$$

Full-Truncation-Euler (FTE). We define an auxiliary sequence $(\tilde{V}_n)_{n=0,\dots,N}$ and the positively truncated sequence $\bar{V}_n := (\tilde{V}_n)^+$ by

$$\tilde{V}_{n+1} = \tilde{V}_n + \kappa(\theta - (\tilde{V}_n)^+) \Delta + \xi \sqrt{(\tilde{V}_n)^+} \Delta W_n^V, \quad \bar{V}_n = (\tilde{V}_n)^+, \quad (6)$$

with $\tilde{V}_0 = v_0$.

For the following result we write $\bar{V}_t^{(N)} := \bar{V}_n$ for $t \in [t_n, t_{n+1})$. The theorem says that, under suitable assumptions, the FTE approximation converges with rate $1/2$:

Theorem 4.1 (Strong FTE rate for CIR, quoted). *Assume that $\nu > 3$ and $2 \leq p < \nu - 1$. Then there exists a constant $C = C(p, T, \kappa, \theta, \xi, v_0)$ such that*

$$\sup_{t \in [0, T]} \|V_t - \bar{V}_t^{(N)}\|_{L^p} \leq C N^{-1/2},$$

where $\bar{V}^{(N)}$ denotes the piecewise constant FTE approximation defined in (6).

Proof. See [2, Theorem 1.1]. □

4.3 Strong Convergence of the Price Process under LE+FTE

The result above only concerns the variance process. For option prices we also need the stock process. Here it is useful to work with the logarithm of the stock price. This gives a simple additive update and keeps the numerical stock price positive after exponentiation.

Remark 4.2 (Log-price representation and heuristic error propagation). *Since the stock price remains positive, we may set $X_t := \log S_t$. Applying Ito's formula gives*

$$dX_t = \left(r - \frac{1}{2}V_t\right) dt + \sqrt{V_t} dW_t^S.$$

Numerically this is useful, because the multiplicative price equation becomes an additive update. Afterwards we return to the price by exponentiation,

$$S_t = e^{X_t},$$

The formula also shows directly where the variance process enters. To see the error propagation, assume heuristically that the variance approximation has rate $1/2$,

$$\sup_{t \in [0, T]} \|V_t - \bar{V}_t^{(N)}\|_{L^2} = O(N^{-1/2}),$$

and considers the difference equation

$$X_t - \widehat{X}_t^{(N)} = -\frac{1}{2} \int_0^t (V_s - \bar{V}_s^{(N)}) ds + \int_0^t (\sqrt{V_s} - \sqrt{\bar{V}_s^{(N)}}) dW_s^S$$

Then Cauchy–Schwarz gives an error of order $O(N^{-1/2})$ for the drift part. For the stochastic part we use Ito's isometry and

$$|\sqrt{a} - \sqrt{b}|^2 \leq |a - b|, \quad a, b \geq 0,$$

This gives the following heuristic estimate for the log-price error:

$$\sup_{t \in [0, T]} \mathbb{E}[|X_t - \widehat{X}_t^{(N)}|^2] = O(N^{-1/2}), \quad \sup_{t \in [0, T]} \|X_t - \widehat{X}_t^{(N)}\|_{L^2} = O(N^{-1/4}).$$

This argument is only meant as intuition. It explains why the logarithmic representation is helpful. The actual theorem used below is stronger and is quoted from the literature.

For the actual simulation we now use this log representation. With the Brownian increments from (5) and the FTE variance $\bar{V}_n$, the log-Euler scheme is

$$\widehat{X}_{n+1} = \widehat{X}_n + \left(r - \frac{1}{2}\bar{V}_n\right)\Delta + \sqrt{\bar{V}_n}\Delta W_n^S, \quad \widehat{X}_0 = \log S_0, \quad (7)$$

and the associated price approximation is

$$\widehat{S}_n := \exp(\widehat{X}_n) \quad (n = 0, \dots, N). \quad (8)$$

For the convergence theorem, let $\widehat{S}^{(N)}$ be the corresponding continuous-time interpolation. The bounds $p^*(\nu)$ and $T^*(p)$ come from the assumptions in the cited theorem. The result is:

Theorem 4.3 (Strong rate for the price process under LE+FTE, quoted). *Assume that the conditions of Cozma and Reisinger are satisfied. Then for all $1 \leq p < p^*(\nu)$ and $T < T^*(p)$,*

$$\sup_{t \in [0, T]} \|S_t - \widehat{S}_t^{(N)}\|_{L^p} \leq C \sqrt{\frac{\log(2N)}{N}},$$

where $\widehat{S}^{(N)}$ is the continuous-time LE+FTE approximation and C depends on $(p, T, S_0, v_0, \kappa, \theta, \xi, \rho)$.

Proof. See [3, Theorem 2.7]. □

4.4 Convergence of the Call Price: Lipschitz Argument and Error Decomposition

We now transfer this to option prices. For a European call with strike $K > 0$ and maturity $T > 0$, write

$$g(s) := (s - K)^+, \quad C_0 := e^{-rT} \mathbb{E}[g(S_T)], \quad C_0^{(N)} := e^{-rT} \mathbb{E}[g(\widehat{S}_T^{(N)})].$$

Then g is 1-Lipschitz on $\mathbb{R}$:

$$|g(s) - g(\tilde{s})| \leq |s - \tilde{s}|.$$

Therefore the time-discretization error is bounded by

$$|C_0 - C_0^{(N)}| = e^{-rT} \left| \mathbb{E}[g(S_T)] - \mathbb{E}[g(\widehat{S}_T^{(N)})] \right| \leq e^{-rT} \mathbb{E}[|S_T - \widehat{S}_T^{(N)}|]. \quad (9)$$

Using the quoted result for the price process with $p = 1$, we get

$$|C_0 - C_0^{(N)}| \leq e^{-rT} C \sqrt{\frac{\log(2N)}{N}}.$$

Monte Carlo estimator. For the Monte Carlo part we simulate M independent paths and define

$$\hat{C}_0^{(M,N)} := \frac{1}{M} \sum_{m=1}^M e^{-rT} g(\hat{S}_T^{(N,m)}).$$

If we set

$$Y_m^{(N)} := e^{-rT} g(\hat{S}_T^{(N,m)}),$$

then the $Y_m^{(N)}$ are independent and have expectation $C_0^{(N)}$. Assuming finite variance, we get

$$\mathbb{E}|\hat{C}_0^{(M,N)} - C_0^{(N)}| \leq \sqrt{\mathbb{E}|\hat{C}_0^{(M,N)} - C_0^{(N)}|^2}$$

By independence,

$$\mathbb{E}|\hat{C}_0^{(M,N)} - C_0^{(N)}|^2 = \text{Var}\left(\frac{1}{M} \sum_{m=1}^M Y_m^{(N)}\right) = \frac{1}{M^2} \sum_{m=1}^M \text{Var}(Y_m^{(N)}) = \frac{\text{Var}(Y_1^{(N)})}{M}.$$

Therefore,

$$\mathbb{E}|\hat{C}_0^{(M,N)} - C_0^{(N)}| \leq \frac{e^{-rT} \sqrt{\text{Var}(g(\hat{S}_T^{(N)}))}}{\sqrt{M}}.$$

4.5 Main Theorem

Theorem 4.4 (Main theorem: total error under LE+FTE+MC). *Under the assumptions of the quoted convergence result for the price process with $p = 1$ and the moment condition*

$$\text{Var}(g(\hat{S}_T^{(N)})) < \infty,$$

there exist constants $C_{\text{disc}}, C_{\text{MC}} \geq 0$ such that

$$\mathbb{E}|C_0 - \hat{C}_0^{(M,N)}| \leq C_{\text{disc}} \sqrt{\frac{\log(2N)}{N}} + C_{\text{MC}} M^{-1/2}.$$

Proof. We prove the estimate by splitting the error. The triangle inequality gives

$$|C_0 - \hat{C}_0^{(M,N)}| \leq |C_0 - C_0^{(N)}| + |C_0^{(N)} - \hat{C}_0^{(M,N)}|.$$

For the first term we use (9) and the strong rate for the price process at $p = 1$:

$$|C_0 - C_0^{(N)}| \leq e^{-rT} \mathbb{E}[|S_T - \hat{S}_T^{(N)}|] \leq C_{\text{disc}} \sqrt{\frac{\log(2N)}{N}}.$$

For the second term we use that $\hat{C}_0^{(M,N)}$ is an average of independent square-integrable random variables. Therefore,

$$\mathbb{E}|C_0^{(N)} - \hat{C}_0^{(M,N)}| \leq \frac{e^{-rT} \sqrt{\text{Var}(g(\hat{S}_T^{(N)}))}}{\sqrt{M}} \leq C_{\text{MC}} M^{-1/2}.$$

Combining the two bounds proves the claim. $\square$

4.6 Calibration Functional

We calibrate the model in the space of implied volatilities, over strikes and maturities. Let

$$\mathcal{E} = \{T_1, \dots, T_J\}, \quad \mathcal{I}_j = \{1, \dots, n_j\}, \quad \mathcal{O} := \{(j, i) : 1 \leq j \leq J, 1 \leq i \leq n_j\}$$

denote the expiries used, the associated strikes, and the full calibration set. The admissible parameter space is

$$\mathcal{P} := (0, \infty)^3 \times (-1, 1) \times [0, \infty).$$

The goal is to find a parameter vector $\Theta = (\kappa, \theta, \xi, \rho, v_0) \in \mathcal{P}$ minimizing the weighted error functional

$$\mathcal{J}(\Theta) := \sum_{(j,i) \in \mathcal{O}} w_{ij} \left(\sigma_{ij}^{\text{model}}(\Theta) - \sigma_{ij}^{\text{mkt}} \right)^2$$

.

The weights w_{ij} are mainly based on the bid-ask spread. A small spread means that the quote is more reliable, so this option receives a larger weight. A large spread receives a smaller weight. In the implementation this is also balanced across moneyness, so that only the at-the-money region does not dominate the whole fit.

Working with implied volatilities makes the smile directly visible. The optimization is performed with L-BFGS-B, a quasi-Newton method with box constraints [1]. We also use restarts and common random numbers. This is done so that the objective values are easier to compare and not dominated by Monte Carlo noise.

5 Data Basis and Calibration Procedure

The numerical part is implemented in Python. The code separates data loading, data cleaning, calibration and plotting. In the following we only describe the steps that are needed to reproduce the figures and tables used in this paper.

5.1 Data Cleaning

The raw data are SPY option quotes from MarketData.app. In the snapshot used here, 4716 raw quotes are loaded. After cleaning, 3469 observations remain. We remove quotes without reliable bid and ask prices, economically implausible contracts, duplicates, violations of simple no-arbitrage bounds and clearly illiquid observations. In the cleaning report, 961 observations are removed because of illiquidity, 246 because of wide spreads and 40 because of arbitrage violations.

For the Heston calibration we use the OTM side for each strike. This avoids an artificial jump when switching between calls and puts. We also restrict the set to $|\log(K/F)| \leq 0.35$. So we focus on the part of the smile where the data quality is still reasonable.

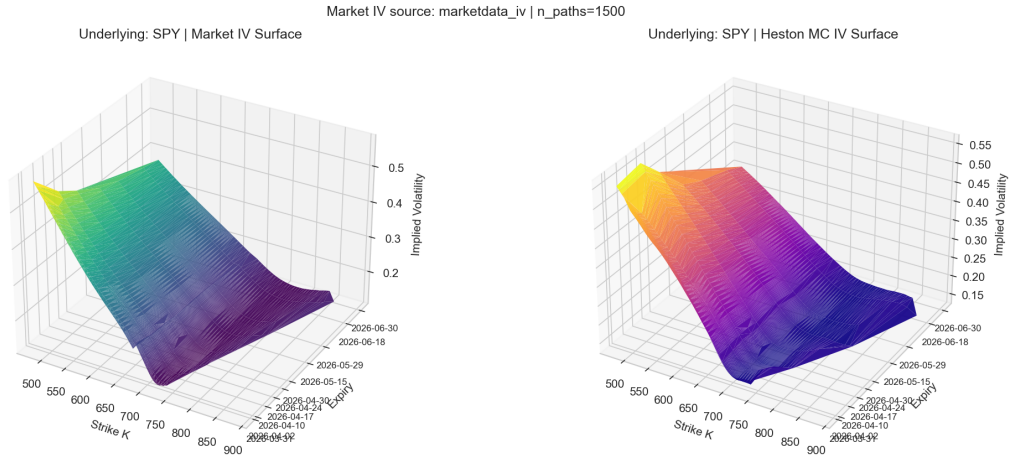


Figure 2: Cleaned implied-volatility surface of the SPY data set used. The calibration does not work on this entire surface directly, but on a consistent sub-universe extracted from it.

The surface is used here mainly to see the structure of the data. The calibration itself uses only a selected subset of this data set.

5.2 Implementation Workflow

The calibration is carried out on three selected expiries:

$$2026-04-17, \quad 2026-06-18, \quad 2026-09-18,$$

The corresponding risk-free rate is determined using bonds with similar maturities; in the reported run, $r = 3.59\%$ is used.

For every parameter candidate Θ we do the same thing. First we simulate variance paths $\hat{V}$. Then we simulate the corresponding stock paths $\hat{S}$. From these paths we compute Monte Carlo prices for the selected options. These prices are then converted

into Black–Scholes implied volatilities. Only after this step are they compared with the market IVs.

For the reported calibration we use 1200 paths per objective evaluation, 140 time steps per year and seed 1234. After filtering and subsampling, 80, 73, and 80 options enter the three maturities. The best candidate is then evaluated once more with 5000 paths.

The optimization uses L-BFGS-B. So the parameter vector should be read as the best point found in this run, not as a mathematically proven global optimum. The full run takes about 710 seconds on my laptop. We reuse the same Gaussian random fields for all candidate parameters. This reduces the Monte Carlo noise when two objective values are compared.

6 Empirical Results

In what follows, IV-RMSE denotes the root mean squared error measured in implied-volatility space and therefore quantifies the average discrepancy between model IV and market IV.

We now look at the numerical results. These results are empirical diagnostics. They do not prove a parameter-specific convergence theorem for the calibrated parameter vector.

Parameter	Value
κ	16.257 477
θ	0.003 705
ξ	4.676 007
ρ	−0.798 064
v_0	0.042 568

Table 1: Calibrated parameters (SPY data set).

The table shows the best parameter vector found in the L-BFGS-B restarts. So it is the best point found by the numerical procedure, not a proven global optimum.

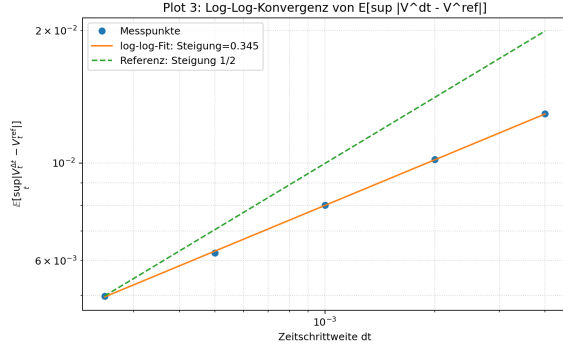
The Feller coefficient

$$\nu = \frac{2\kappa\theta}{\xi^2} \approx 0.00551$$

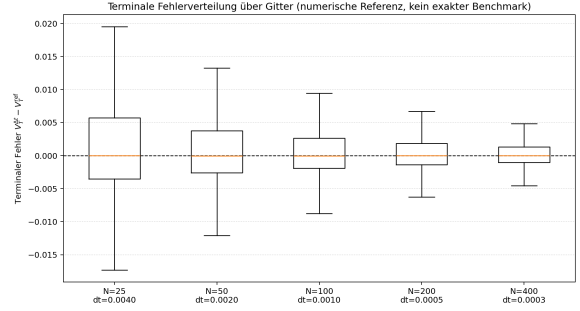
lies far below 3. The result of Cozma and Reisinger cited in Chapter 4 therefore does not cover this parameter vector. All subsequent convergence and stability statements are thus explicitly numerical diagnostics rather than convergence proofs for this specific parameter choice.

To assess the numerical plausibility of the calibrated parameter vector nonetheless, a separate test against a very finely discretized reference solution was carried out. Here the error of the implemented scheme is not measured against the unknown exact solution, but

against a highly resolved reference simulation. Such a diagnostic cannot prove convergence in the strict sense; it can only indicate whether the relevant computational range exhibits a stable error profile that decreases under refinement. It is noteworthy that [2] also report stable behavior in numerical experiments for parameter constellations below the theoretical threshold $\nu > 3$.



(a) Supremum error against the reference solution

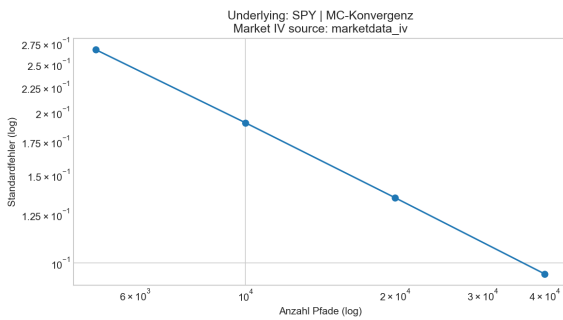


(b) Terminal error relative to the reference solution

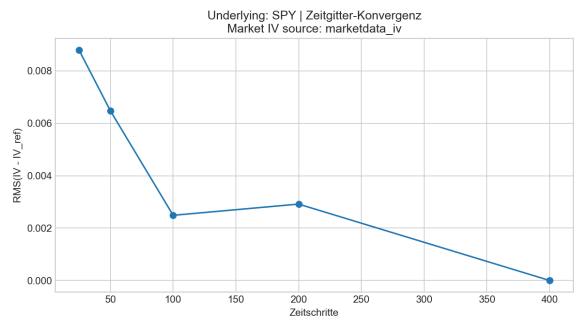
Figure 3: Heuristic stability diagnostics of the FTE scheme for the calibrated parameter vector, compared with a highly resolved reference solution.

The two figures show that the error decreases when the grid is refined. This is exactly what one expects from a stable numerical scheme. Still, this is only a numerical check and not a proof of convergence.

6.1 Convergence Diagnostics



(a) Monte Carlo convergence



(b) Δt -convergence

Figure 4: Heuristic convergence diagnostics for sample size and time step.

The diagnostics are based on a separate validation run with seed 20260309, path-count grid (500, 1000, 2000, 5000, 10000, 20000), and step-count grid (10, 20, 50, 100, 200, 500). For the Monte Carlo curve, a log-log regression of the standard errors against the number of paths yields a slope of -0.509845 , which is consistent with the theoretical rate $M^{-1/2}$.

For the time-step study, the maximal relative deviation from the finest grid with 500 steps is 0.03498; at 200 steps it decreases to 0.005885. In addition, comparison with a high-precision reference Monte Carlo valuation gives a call-price RMSE of 0.131057 and an IV-RMSE of 0.006908.

These figures do not establish parameter-specific strong convergence of the implemented scheme. They merely show that, in the computational range considered, there are no obvious instabilities and no behavior fundamentally at odds with the expected error logic.

6.2 Martingale Property of the Discounted Price Process

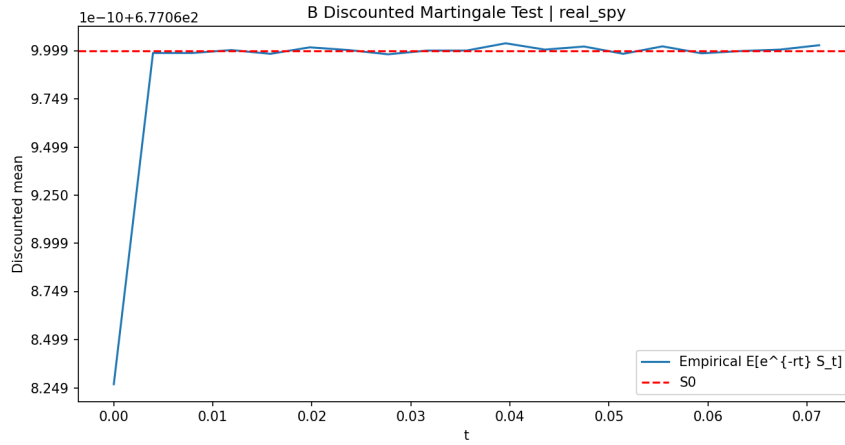


Figure 5: Discounted mean $E[e^{-rt}S_t]$ in the validation test compared with S_0 .

Under the risk-neutral measure the discounted price process $e^{-rt}S_t$ should be a martingale. Numerically we check this by plotting the simulated mean of $e^{-rt}S_t$ over time and comparing it with S_0 .

This is only a consistency check. It is not a proof that the implementation is correct. In the validation report the maximal relative deviation is $2.56 \cdot 10^{-13}$. So in this run there is no visible contradiction to the martingale property.

6.3 In-Sample and Out-of-Sample Fit

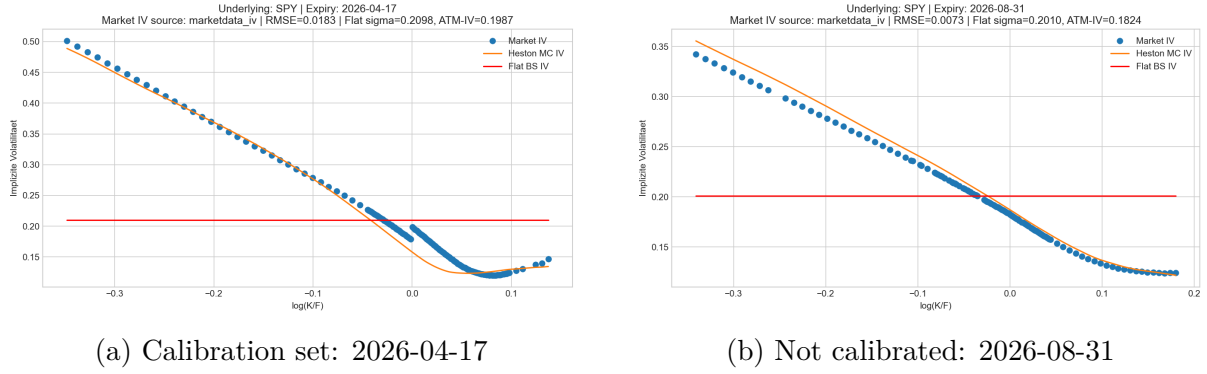


Figure 6: Smile comparison for one calibrated expiry and one expiry not used in the calibration.

For the calibrated expiry the model follows the market smile quite well in the central moneyness region. The largest errors are in the wings. For the expiry that was not used in the calibration, the same general pattern appears. The center is still reasonable, while the outer regions are harder to fit. This comparison should therefore be read as a diagnostic plot and not as a separate statistical validation.

6.4 Residual Structure

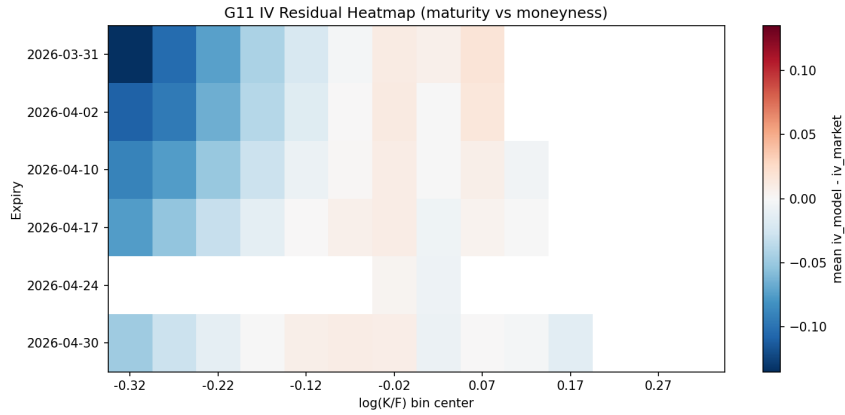


Figure 7: Residual heatmap (model IV minus market IV) over maturity and moneyness.

For 805 market points the residual diagnostics give an IV-RMSE of 0.032212 and a price-RMSE of 0.636885. The heatmap shows that the errors are not randomly scattered around zero. They have a visible structure over maturity and moneyness. This can be interpreted as a limitation of the model. However, data filtering, weighting choices and remaining Monte Carlo noise may also contribute.

7 Model Limitations and Outlook toward Rough Volatility

The residuals suggest that the classical Heston model may be too rigid. It has only one Markovian volatility factor. Therefore it cannot always fit the whole surface over moneyness and maturity at the same time. This is why we briefly look at rough-volatility models.

Typical restrictions are:

- a Markov structure with only one volatility factor,
- limited flexibility, especially for short maturities,
- systematic residual patterns when fitting several expiries simultaneously.

Rough-volatility models address these deficits by describing the variance no longer through a purely Markovian dynamics, but through a Volterra kernel with memory:

$$V_t = V_0 + \int_0^t K(t-s)\kappa(\theta - V_s) ds + \int_0^t K(t-s)\xi\sqrt{V_s} dW_s,$$

with

$$K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}, \quad H \in (0, 1/2).$$

The case $H < 1/2$ corresponds to the empirically observed roughness of volatility, that is, a highly irregular short-term dynamics [5, 4]. Compared with the classical Heston framework, the memory structure opens up additional degrees of freedom for a simultaneous fit across moneyness and maturity. In this sense, rough-volatility models form a natural extension class if the residual patterns visible in the present data set cannot already be explained by numerical or data-related effects.

The aim here is not to develop rough Heston in detail. We only use it to explain why the remaining residual patterns are plausible from a modelling point of view.

8 Conclusion

We have now seen the main mathematical and numerical points of the Heston model. The CIR variance process has a well-defined strong solution, remains non-negative, and has a boundary behavior described by the Feller condition. For the numerical part we used the full-truncation Euler scheme for the variance and the log-Euler scheme for the stock price. The total error was split into a discretization part and a Monte Carlo part.

In the empirical part the Heston model reproduces the central part of the SPY volatility smile reasonably well. At the same time, the calibrated parameters are outside the

range covered by the quoted FTE theorem. Therefore the numerical tests should be read as checks and not as proofs. The remaining structured residuals show that the model is useful, but not a final description of the data.

A Auxiliary Results for Chapter 3

Theorem A.1 (Deterministic time change). *Let W be a Brownian motion and let $\tau : [0, \infty) \rightarrow [0, \infty)$ be deterministic, C^1 , strictly increasing and satisfy $\tau(0) = 0$. Then there exists a Brownian motion $\widehat{W}$ such that, for suitable integrands H ,*

$$\int_0^t H_s dW_{\tau(s)} = \int_0^t H_s \sqrt{\tau'(s)} d\widehat{W}_s.$$

In differential notation this means

$$dW_{\tau(s)} = \sqrt{\tau'(s)} d\widehat{W}_s.$$

This is the standard deterministic time-change rule for continuous martingales; see [8, 6].

Lemma A.2 (Product rule with a deterministic factor). *Let X be a continuous semimartingale and let $f \in C^1([0, \infty))$ be deterministic. Then*

$$d(f(t)X_t) = f(t) dX_t + f'(t)X_t dt.$$

This is the usual product rule for semimartingales; see for example [6].

Lemma A.3 (Normalization of a continuous martingale). *Let $(M_t)_{t \geq 0}$ be a continuous local martingale with $\langle M \rangle_t = ct$ for some $c > 0$. Then $\widetilde{W}_t := M_t/\sqrt{c}$ is a Brownian motion.*

This follows from Levy's characterization of Brownian motion; see [8].

Definition A.4 (Pathwise uniqueness). *An SDE has pathwise uniqueness if, on every probability space with the same Brownian motion and the same initial value, two arbitrary solutions X^1 and X^2 are indistinguishable, that is,*

$$\mathbb{P}(X_t^1 = X_t^2 \text{ for all } t \geq 0) = 1.$$

See [9] for the Yamada–Watanabe principle.

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